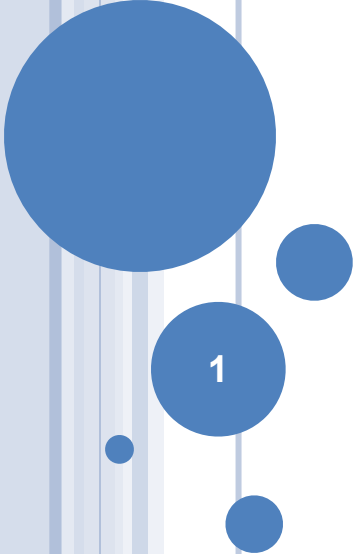


INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION



Chapter 5 Several Useful Discrete Distributions

1

A decorative graphic on the left side of the slide consists of several vertical stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle is particularly large and prominent, while others are smaller and scattered around it.

5.1 DISCRETE RVs AND THEIR PROBABILITY DISTRIBUTIONS

2

KEY IDEAS

- Random variable
- Discrete random variables
- Probability distribution
- The calculation of mean, variance, standard deviation.

PROBABILITY DISTRIBUTIONS

- Chapters 1 and 2: Relative frequency
- Chapter 4: Probability
- Chapter 5 and 6: Discrete and Continuous Random Variables

We define the probability distribution for a random variable X as the relative frequency distribution constructed for the entire population of measurements.

RANDOM VARIABLES

- A **variable** X is a **random variable** if the value that it assumes, corresponding to the outcome of an experiment, is a chance or random event.
- It can be **discrete** and **continuous**.

Examples

- X : Number of defects on a randomly selected piece of furniture
- X : SAT score for a randomly selected college applicant
- X : Number of telephone calls received by a crisis intervention hotline during a randomly selected time period

PROBABILITY DISTRIBUTIONS FOR DISCRETE RANDOM VARIABLES

- The **probability distribution for a discrete random variable x** resembles the relative frequency distributions we constructed in Chapter 1.
- It is a graph, table or formula that gives the possible values of x and the probability $p(x)$ associated with each value.

We must have

$$0 \leq p(x) \leq 1 \text{ and } \sum p(x) = 1$$

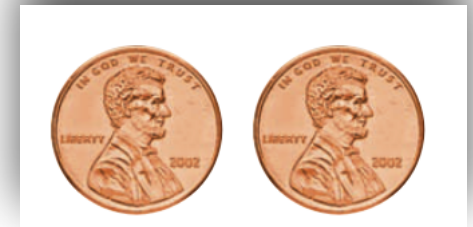
Here, $p(x)$ is called the frequency function, the **probability mass function (pmf)**, or **probability distribution**.

EXAMPLE 5.1

- Toss two fair coins and let X equal the number of heads observed. Find the probability distribution for X .

■ **Table 5.1 Simple Events and Probabilities in Tossing Two Coins**

Simple Event	Coin 1	Coin 2	$P(E_i)$	x
E_1	H	H	1/4	2
E_2	H	T	1/4	1
E_3	T	H	1/4	1
E_4	T	T	1/4	0

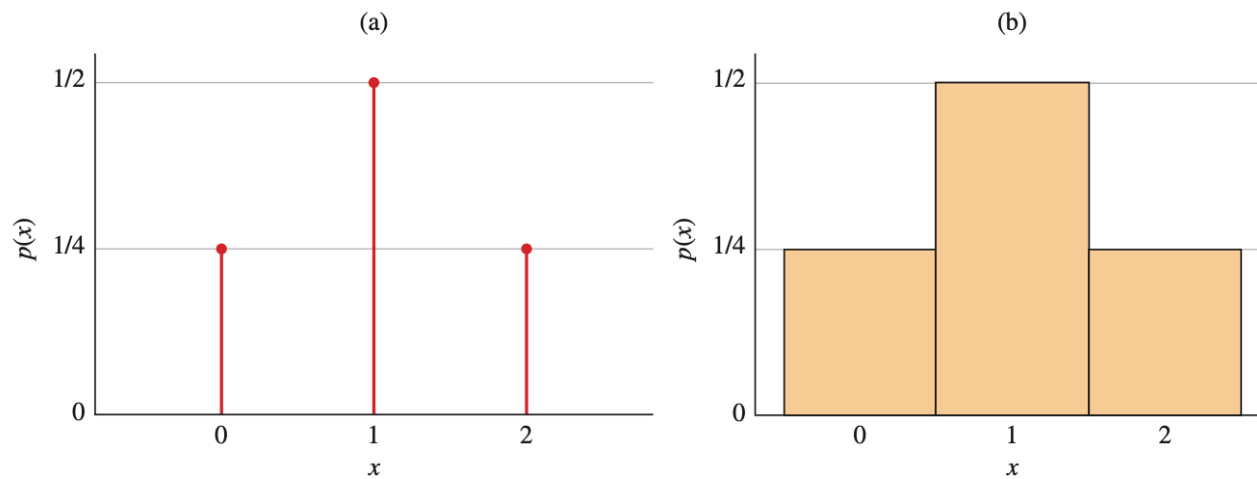


■ **Table 5.2 Probability Distribution for x (x = Number of Heads)**

x	Simple Events in x	$p(x)$
0	E_4	1/4
1	E_2, E_3	1/2
2	E_1	1/4
$\Sigma p(x) = 1$		

Figure 5.1

Probability histograms for Example 5.1



EXAMPLE

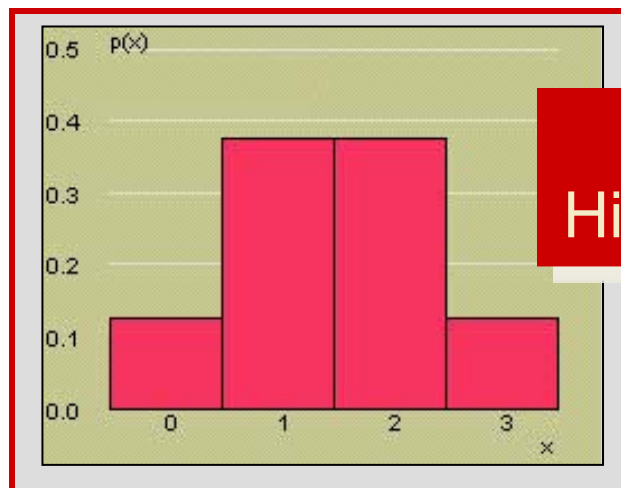
- Toss a fair coin three times and define $x = \text{number of heads}$.



		<u>x</u>
HHH	1/8	3
HHT	1/8	2
HTH	1/8	2
THH	1/8	2
HTT	1/8	1
THT	1/8	1
TTH	1/8	1
TTT	1/8	0

$$\begin{aligned}
 P(x = 0) &= 1/8 \\
 P(x = 1) &= 3/8 \\
 P(x = 2) &= 3/8 \\
 P(x = 3) &= 1/8
 \end{aligned}$$

x	$p(x)$
0	1/8
1	3/8
2	3/8
3	1/8



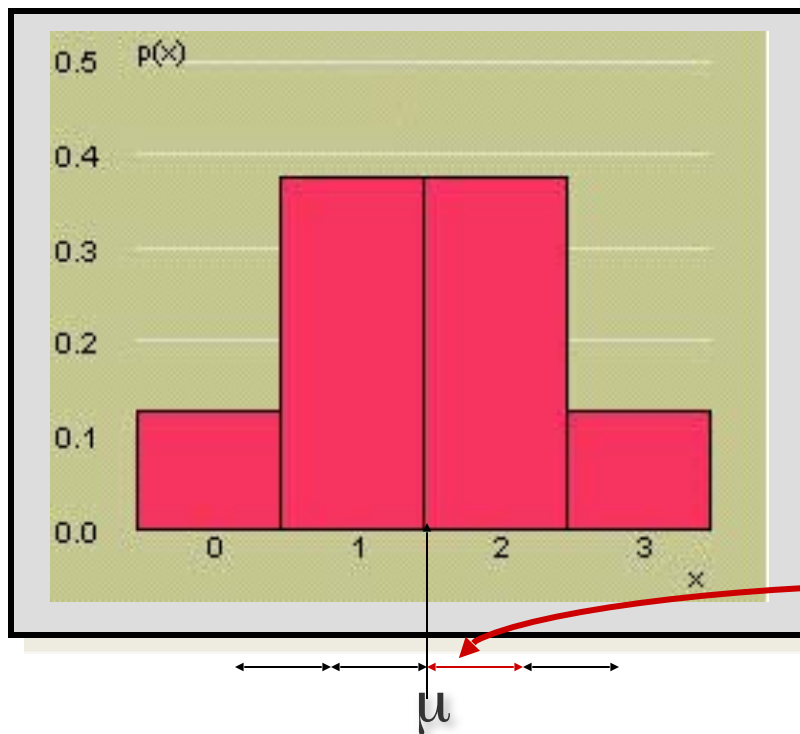
Probability
Histogram for x

PROBABILITY DISTRIBUTIONS

- Probability distributions can be used to describe the population, just as we described samples in Chapter 1.
 - **Shape:** Symmetric, skewed, mound-shaped...
 - **Outliers:** unusual or unlikely measurements
 - **Center and spread:** mean and standard deviation. A population mean is called μ and a population standard deviation is called σ .

EXAMPLE

- The probability distribution for x the number of heads in tossing 3 fair coins.



- Shape?
- Outliers?
- Center?
- Spread?

Symmetric;
mound-shaped

None

$\mu = 1.5$

$\sigma = .688$

THE MEAN AND STANDARD DEVIATION

- Let X be a discrete random variable with probability distribution $p(x)$.
- The **mean** or **expected value** of X is

$$\mu = E[X] = \sum xp(x)$$

- The **variance** of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum (x - \mu)^2 p(x)$$

- The **standard deviation** of X is

$$\sigma = \sqrt{\sigma^2} = \sqrt{\sum (x - \mu)^2 p(x)}$$

EXAMPLE



- Toss a fair coin 3 times and record x the number of heads.

x	$p(x)$	$xp(x)$	$(x-\mu)^2p(x)$
0	$1/8$	0	$(-1.5)^2(1/8)$
1	$3/8$	$3/8$	$(-0.5)^2(3/8)$
2	$3/8$	$6/8$	$(0.5)^2(3/8)$
3	$1/8$	$3/8$	$(1.5)^2(1/8)$

$$\mu = \sum xp(x) = \frac{12}{8} = 1.5$$

$$\sigma^2 = \sum (x - \mu)^2 p(x)$$

$$\sigma^2 = .28125 + .09375 + .09375 + .28125 = .75$$

$$\sigma = \sqrt{.75} = .688$$

FIND μ AND σ OF EXAMPLE 5.1

- $\mu = 0 \times 1/4 + 1 \times 1/2 + 2 \times 1/4 = 1$
- $\sigma^2 = (0 - 1)^2 \times 1/4 + (1 - 1)^2 \times 1/4 + (2 - 1)^2 \times 1/4 = 1/2$
- $\sigma = \sqrt{1/2} = 0.71$

x	0	1	2
$p(x)$	1/4	1/2	1/4

DISCRETE RVs

- Discrete random variables take on only a finite or countably infinite number of values.
- Three discrete probability distributions serve as models for a large number of practical applications:

✓ Ch 5.2: The **binomial** random variable

✓ Ch 5.3: The **Poisson** random variable

✓ Ch 5.4: The **hypergeometric** random variable

The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, and several smaller circles are arranged vertically below it. The title '5.2 BINOMIAL DISTRIBUTION' is written in a white, serif font, positioned to the right of the circles.

5.2 BINOMIAL DISTRIBUTION

16

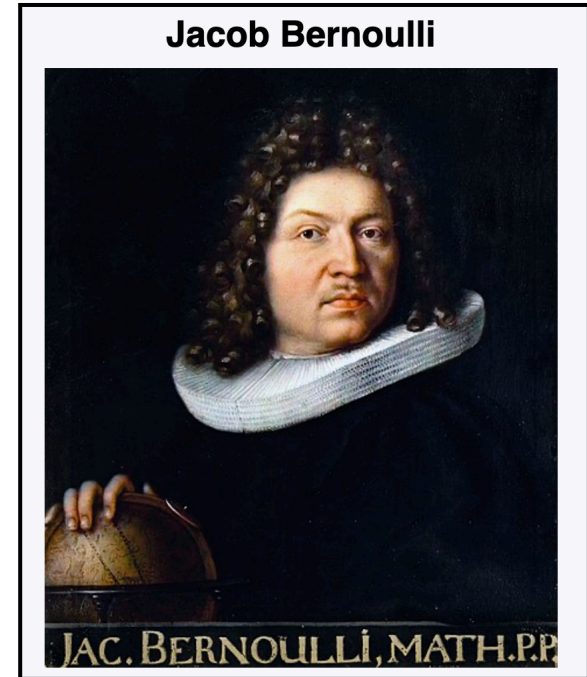
THE BERNOULLI TRAIL

1. Each trial results in **one of two outcomes**, success (S) or failure (F).
2. The probability of success on a single trial is p and **remains constant** from trial to trial. The probability of failure is $q = 1 - p$.
3. Bernoulli random variable defines X as 1 for a success and 0 for a failure in a Bernoulli trial, denoted as $X \sim \text{Bernoulli}(p)$. Its probability distribution is

$$p(1) = p, \quad p(0) = 1 - p.$$

Or,

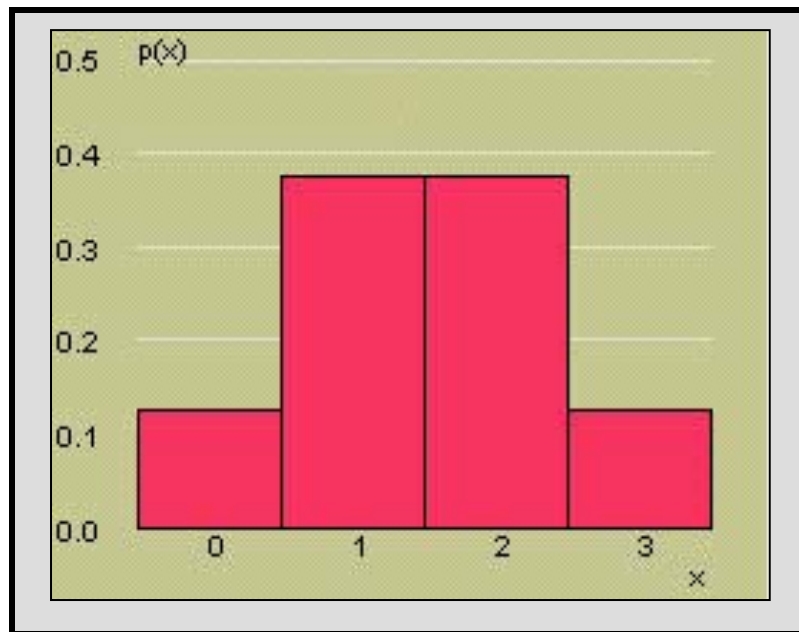
$$p(x) = p^x(1 - p)^{1-x}, \quad \text{for } x = 0, 1.$$



1655-1705, Swiss Mathematician

THE BINOMIAL RANDOM VARIABLE

- The **coin-tossing experiment** is a simple example of a **binomial random variable**. Toss a fair coin $n = 3$ times and record $x =$ number of heads.

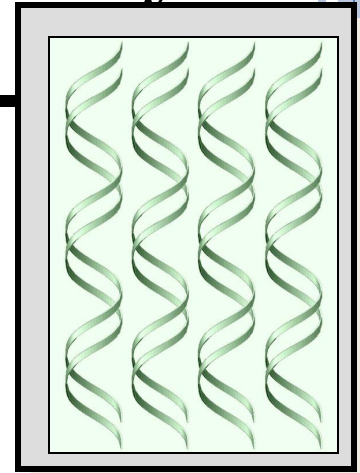


x	$p(x)$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$

THE BINOMIAL RANDOM VARIABLE

- Many situations in real life resemble the coin toss, but the coin is not necessarily fair, so that $P(H) \neq 1/2$.

- Example:** A geneticist samples 10 people and counts the number who have a gene linked to Alzheimer's disease.



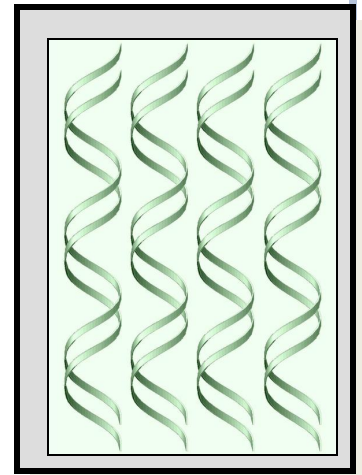
- | | | | |
|----------------|-------------------|--------------------|--|
| • Coin: | Person | • Number of | $n = 10$ |
| • Head: | Has gene | tosses: | P(has gene) =
proportion in the
population who have
the gene. |
| • Tail: | Doesn't have gene | • P(H): | |

THE BINOMIAL EXPERIMENT

1. The experiment consists of n identical **Bernoulli trials**.
2. Each trial results in **one of two outcomes**, success (S) or failure (F).
3. The probability of success on a single trial is p and **remains constant** from trial to trial. The probability of failure is $q = 1 - p$.
4. The trials are **independent**.
5. We are interested in X , the number of **successes in n trials**.

BINOMIAL OR NOT?

- Very few real life applications satisfy these requirements exactly.



-
- Select two people from the U.S. population, and suppose that 15% of the population has the Alzheimer's gene.
 - For the first person, $p = P(\text{gene}) = .15$
 - For the second person, $p \approx P(\text{gene}) = .15$, even though one person has been removed from the population.

THE BINOMIAL PROBABILITY DISTRIBUTION

- For a binomial experiment with n trials and probability p of success on a given trial, the probability of x successes in n trials is

$$P(X = x) = C_x^n p^x (1 - p)^{(n-x)}, \text{ for } x = 0, 1, \dots, n.$$

$$C_x^n = \frac{n!}{x!(n-x)!},$$

with $n! = n(n-1)\cdots(2)(1)$, **and** $0! = 1$

$$\mu = \sum_{x=0}^n x C_x^n p^x (1-p)^{(n-x)} = np.$$

THE MEAN AND STANDARD DEVIATION

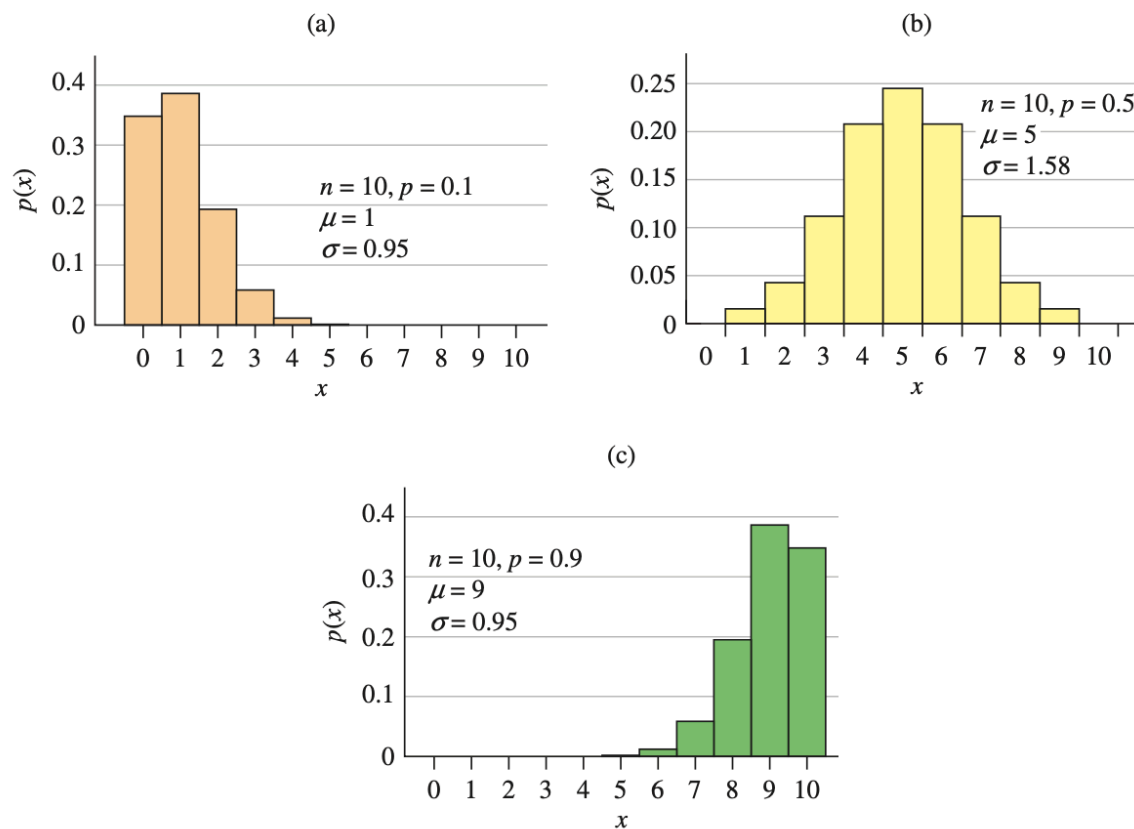
- For a binomial experiment with n trials and probability p of success on a given trial, the measures of center and spread are:

Mean: $\mu = np$

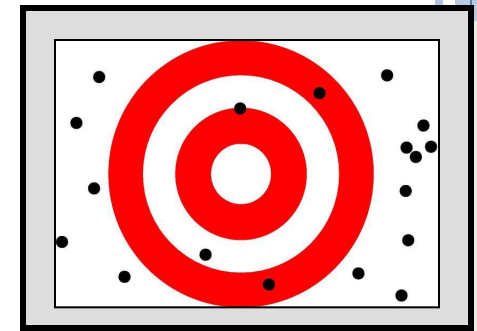
Variance: $\sigma^2 = npq$

Standard deviation: $\sigma = \sqrt{npq}$

Fig 5.3 Binomial Probability Distributions



EXAMPLE



A marksman hits a target 80% of the time. He fires five shots at the target. What is the probability that exactly 3 shots hit the target?

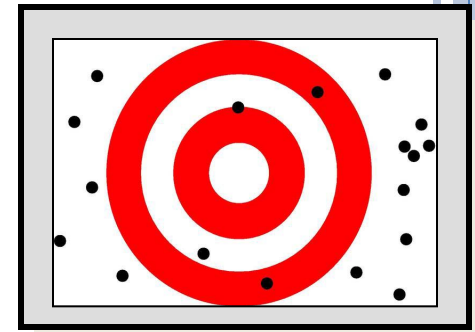
$X \sim \text{Binomial}(n=5, p=0.8)$

$n = 5$ $\text{success} = \text{hit}$ $p = .8$ $x = \text{\# of hits}$

$$P(x = 3) = C_3^5 p^3 q^{5-3} = \frac{5!}{3!2!} (.8)^3 (.2)^{5-3}$$

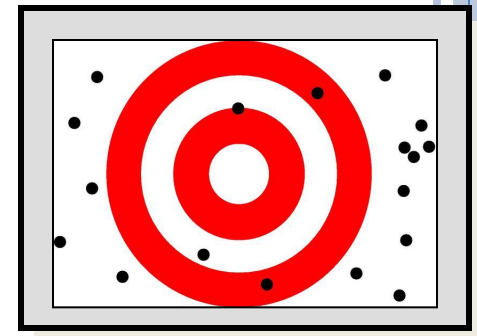
$$= 10(.8)^3 (.2)^2 = .2048$$

EXAMPLE



What is the probability that more than 3 shots hit the target?

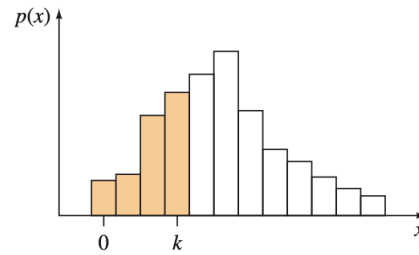
$$\begin{aligned} P(x > 3) &= C_4^5 p^4 q^{5-4} + C_5^5 p^5 q^{5-5} \\ &= \frac{5!}{4!1!} (.8)^4 (.2)^1 + \frac{5!}{5!0!} (.8)^5 (.2)^0 \\ &= 5(.8)^4 (.2) + (.8)^5 = .7373 \end{aligned}$$



CUMULATIVE PROBABILITY TABLES

You can use the **cumulative probability tables** to find probabilities for selected binomial distributions.

- ✓ Find the table for the correct value of ***n***.
- ✓ Find the column for the correct value of ***p***.
- ✓ The row marked “***k***” gives the cumulative probability, $P(x \leq k) = P(x = 0) + \dots + P(x = k)$ ²⁷



■ **Table 1 Cumulative Binomial Probabilities**

Tabulated values are $P(x \leq k) = p(0) + p(1) + \dots + p(k)$.
(Computations are rounded to the third decimal place.)

$n = 2$

p														
k	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99	k
0	.980	.902	.810	.640	.490	.360	.250	.160	.090	.040	.010	.002	.000	0
1	1.000	.998	.990	.960	.910	.840	.750	.640	.510	.360	.190	.098	.020	1
2	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	2

$n = 3$

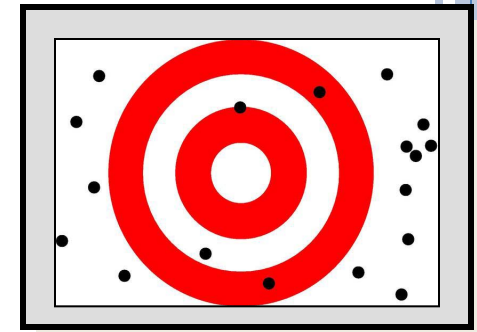
	p													
k	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99	k
0	.970	.857	.729	.512	.343	.216	.125	.064	.027	.008	.001	.000	.000	0
1	1.000	.993	.972	.896	.784	.648	.500	.352	.216	.104	.028	.007	.000	1
2	1.000	1.000	.999	.992	.973	.936	.875	.784	.657	.488	.271	.143	.030	2
3	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	3


$$\begin{aligned} \mathbf{P(x = 3)} &= P(x \leq 3) - P(x \leq 2) \\ &= .263 - .058 \\ &= .205 \end{aligned}$$

Check from formula:
 $P(x = 3) = .2048$

[illegible]

EXAMPLE



k	$p = .80$
0	.000
1	.007
2	.058
3	.263
4	.672
5	1.000

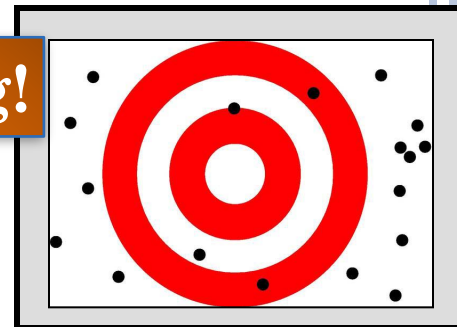
What is the probability that more than 3 shots hit the target?

$$\begin{aligned} P(x > 3) &= 1 - P(x \leq 3) \\ &= 1 - .263 = .737 \end{aligned}$$

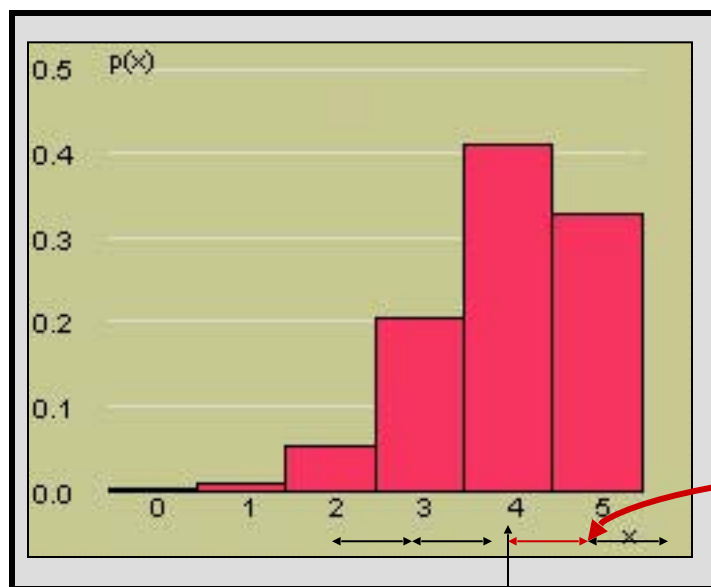
Check from formula:
 $P(x > 3) = .7373$

EXAMPLE

Link to Hypothesis Testing!



- Here is the probability distribution for $x = \text{number of hits}$. What are the mean and standard deviation for x ?



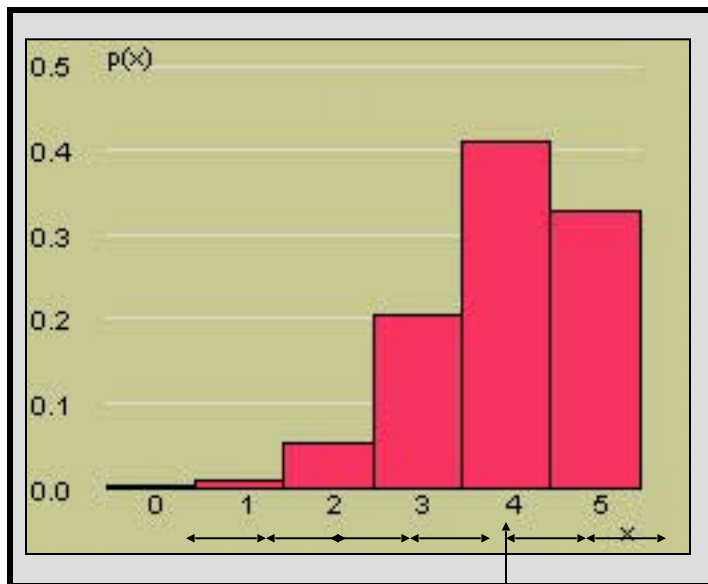
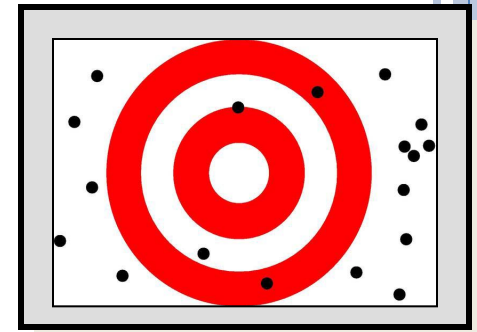
$$\text{Mean : } \mu = np = 5(.8) = 4$$

$$\begin{aligned}\text{Standard deviation: } \sigma &= \sqrt{npq} \\ &= \sqrt{5(.8)(.2)} = .89\end{aligned}$$

μ

Link to Hypothesis Testing!

- Would it be unusual to find that none of the shots hit the target?



μ

The value:

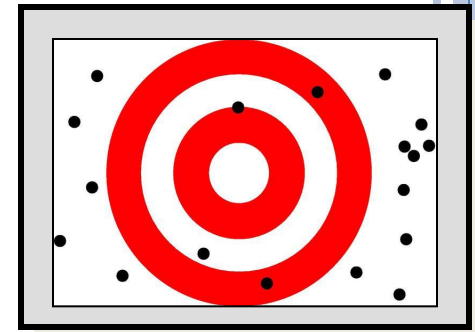
$$P(x = 0) = 0.2^5 = 0.00032$$

This is very unusual.

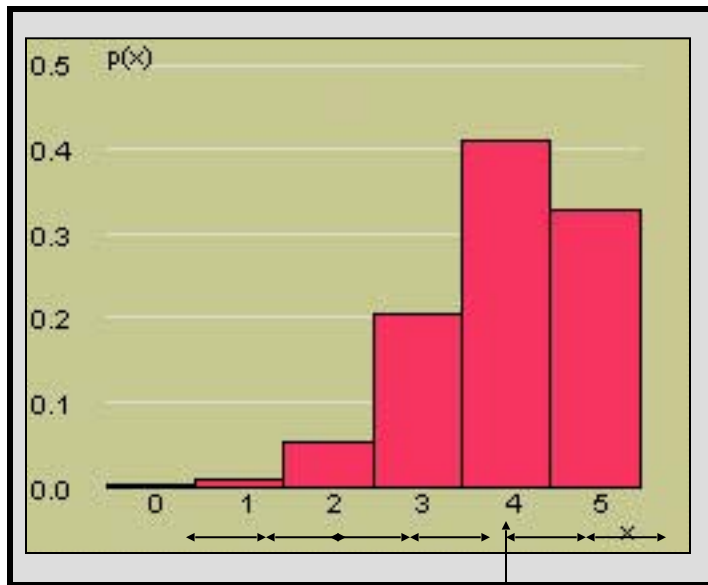
Link to Hypothesis Testing!

EXAMPLE

- Would it be unusual to find that none of the shots hit the target?



$$\mu = 4; \sigma = .89$$



μ

- The value $x = 0$ lies
$$z = \frac{x - \mu}{\sigma} = \frac{0 - 4}{.89} = -4.49$$
- more than 4 standard deviations below the mean. Very unusual.

The left side of the slide features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several blue circles of different sizes. One large circle is positioned near the top left, and several smaller circles are scattered below it, some overlapping the stripes.

SUMMARY

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KEY CONCEPTS

I. The Binomial Random Variable

1. Five characteristics: n identical independent trials, each resulting in either success S or failure F ; probability of success is p and remains constant from trial to trial; and x is the number of successes in n trials.

2. Calculating binomial probabilities

a. Formula: $P(x = k) = C_k^n p^k q^{n-k}$

b. Cumulative binomial tables

c. Individual and cumulative probabilities using Minitab

3. Mean of the binomial random variable: $\mu = np$

4. Variance and standard deviation: $\sigma^2 = npq$ and

$$\sigma = \sqrt{npq}$$